INCH-POUND

MIL-H-14516F
28 July 1993
SUPERSEDING
MIL-H-14516E
30 December 1987

MILITARY SPECIFICATION

HEATER, SPACE: OIL FIRED, BARRACKS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one size of oil-fired, general-purpose space heater having a heat output of 60,000 BTU/hr at 0.060 inch water gauge draft. The space heater utilizes a standard military rectangular 5 gallon fuel can as a fuel tank.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1898 - Cushioning Material, Cellulosic, Packaging
FF-N-105 - Nails, Brads, Staples and Spikes: Wire, Cut and Wrought

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4520

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UU-T-81	-	Tags, Shipping and Stock
VV-F-800	-	Fuel Oil, Diesel
PPP-B-591	-	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-60	-	Tape: Packaging, Waterproof
PPP-T-97	-	Tape, Packaging/Industrial, Filament Reinforced

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-L-21260	-	Lubricating Oil, Internal Combustion Engine, Preservative and Break-In

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-248	-	Welding and Brazing Procedure and Performance Qualification
MS 24629	-	Screw, Tapping-Thread Cutting, Types D, F, G, or T Pan Head, Cross-Recessed, Steel, Carbon, Cadmium Plated
MS 27183	-	Washer, Flat (Round, Steel, Cadmium Plated) General Purpose
MS 35206	-	Screw, Machine-Pan Head, Cross-Recessed, Carbon Steel, Cadmium Plated, UNC-2A
MS 35338	-	Washer, Lock-Spring, Helical, Regular (Medium) Series

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-13-2851	-	Heater, Space, Oil-Fired, 70,000 BTU/hr; Drawing and Parts List
13208 E 6211	-	Valve, Constant Level

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-UXT, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 641M	-	Zinc-Coated (Galvanized) Carbon Steel Wire
ASTM D 2156	-	Smoke Density in Flue Gases from Burning Distillate Fuels
ASTM D 3953	-	Strapping, Flat Steel and Seals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 507	-	Electric Fans
UL 896	-	Oil-Burning Stoves

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR-CONDITIONING ENGINEERS (ASHRAE) - AMCA STANDARD

ASHRAE 51-1985/AMCA 210-85 - Laboratory Methods of Testing Fans for Ratings

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(Applications for copies should be addressed to Air Moving and Conditioning, Inc., 30 West University Drive, Arlington Heights, IL 60004, or American Society of Heating, Refrigerating and Air-Conditioning Engineers, 1791 Tullie Circle, N.E., Atlanta, GA 30329.)

AMERICAN WELDING SOCIETY (AWS)

AWS Standard B3.0 - Welding Procedure and Performance Qualification

(Application for copies should be addressed to the American Welding Society, 550 NW Lejeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, except for related associated detail specifications, specification sheets, or MS standards, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Codes and standards. The space heaters shall conform to the applicable requirements of Underwriters Laboratories (UL) Standards 507 and 896.

3.3 Materials and components. Materials and components shall be as specified herein and as shown on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the contractor in the manufacture of his space heater, provided the completed item complies with all provisions of this document. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Fan. The construction of the fan shall be in accordance with UL Standard 507 (see 4.4.1.2).

3.3.1.1 Fan capacity. The fan shall be tested as specified in 4.4.5 to determine fan capacity for 0 pressure.

3.4 Design and construction. The heater shall be in accordance with drawings listed on Drawing 5-13-2851 and as specified herein. The drawings forming a part of this document are engineering design drawings. No

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deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Unless otherwise specified in the contract or purchase order, any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request for inspection by the contracting officer or his designated representative (see 6.2). When specified (see 6.2), a smoke jack assembly and a fan assembly shall be furnished. The heater shall have a service life of 3,000 hours before requiring repairs or replacement of parts, shall operate for a minimum of 300 hours prior to cleaning the burner pot, and shall operate for 600 hours without requiring maintenance other than cleaning of the burner pot each 300 hours. Testing for service life shall be as specified in 4.4.5.

3.4.1 Heater output. The heaters shall develop the heat outputs specified in table I, without visible smoke, when burning grade DF-2 Continental U.S. diesel fuel conforming to VV-F-800. Testing shall be as specified in 4.4.5. Output and performance shall also conform to UL Standard 896 except for fuel flow and control valve calibration.

TABLE I. Heat outputs

BTU/hr	Fuel rate (cubic centimeters/minute at 4.00 centistokes at 70°F)	Draft (inches of water)
60,000 (minimum high-fire)	48 ± 2	0.060 ± 0.005
20,000 (maximum low-fire)	6 ± 1	0.060 ± 0.005

3.4.2 Seepage and burn-off. The heater shall conform to the seepage and burn-off requirements specified in UL Standard 896. Testing shall be as specified in 4.4.5.

3.4.3 Tilted operation. A tilt of 2 degrees in any direction from horizontal shall not affect the starting or specified operation of the heater. Testing shall be as specified in 4.4.5.

3.4.4 Fuel control valve flow rate. The flow rate of the fuel control valve component shall be as specified on Drawing 13208 E 6211 (see 4.4.1.2).

3.4.5 Siphon tube and fuel control valve assembly flow rate. When tested in accordance with 4.4.5, the fuel shall flow freely when the valve is set at each of the "open" position settings and shall stop flowing when the valve is set in the "off" position.

3.4.6 Smoke reading. When operated at the high-fire fuel rate, the smoke reading shall be less than the No. 6 smoke spot number of ASTM D 2156 when tested as specified in 4.4.5.

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3.4.7 Combustion efficiency. When operated at the high-fire fuel rate and at an acceptable smoke reading, the combustion efficiency shall be greater than 60 percent when tested as specified in 4.4.5. The efficiency shall be determined by the carbon dioxide (CO_2) and the temperature increase of the flue gas.

3.5 Marking for identification. Heaters shall be marked for identification in accordance with the unit marking requirements of MIL-STD-130 and as specified on the applicable drawings.

3.6 Data nameplates. The space heater shall be furnished with a data nameplate in accordance with the applicable requirements of MIL-STD-130 except the requirements for: (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data nameplates shall be made of minimum 20 gauge corrosion-resisting metal and attached to the space heater by rivets, screws, or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The nameplate shall contain the following information which shall be stamped, engraved, or applied by photosensitive means:

National stock number
Procurement Instrument Identification Number
Specification data
Manufacturer's name, address, and telephone number
Supplier's name, address, and telephone number (list only if different from manufacturer)
Manufacturer's model number
Government approved manual number (see 6.2)

Each nameplate shall be placed so that it is readily visible to the operator during normal operating use, and so as to not adversely affect the life and utility of the space heater.

3.7 Workmanship.

3.7.1 Metal fabrication. Sheet metal and structural members used in fabrication shall be straight and free from laminations. Sheet metal and structural members may be cut by shearing or sawing. All burrs and rough edges shall be removed. Corners shall be square and true. All bends shall be made to a radius which will not cause cracking at the extreme fibers of the piece being bent, and all bends except for minor details shall be made by use of metal dies or fixtures to insure uniformity in size and shape.

3.7.2 Bolted connections. Bolt holds shall be formed or drilled and shall have the burrs removed. Washers and lockwashers shall be provided, and all bolts, nuts, and screws shall be tightened so that they will not work loose under vibration.

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3.7.3 Riveted connections. Rivet holes shall be punched or drilled and shall have the burrs removed. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surface of the members.

3.7.4 Welders and welding.

3.7.4.1 Welders. All welders shall maintain current qualifications appropriate to their job under one of the following qualification standards:

(a) Welding Procedure and Performance Qualification of the American Welding Society AWS Standard B3.0
(b) ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications
(c) MIL-STD-248

3.7.4.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be in accordance with AWS Standard B3.0.

3.7.5 Jigs and fixtures. Shop-fabricated components shall be assembled in steel jigs or frames and joined while held in position. Jigs or frames shall minimize distortion.

3.7.6 Interchangeability. All like components shall be interchangeable with respect to installation, performance, and quality.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in 4.4.5. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Parts and components. Parts and components detailed on drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings (see 6.4). The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection.

4.4.1.2 Components and material certification. A certificate of compliance indicating the fuel control valve flow rate meets the requirements of 3.4.4 and that the fan construction meets the requirement of 3.3.1 shall be acceptable.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

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4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Color not as specified	101	
	Not smoothly applied, blisters, flaking, peeling, or scratches		201
Construction and workmanship	Component missing, not in proper location, or not as specified	102	
	Fabrication not as specified	103	
	Bolted and riveted connections not as specified	104	
	Washers or lockwashers missing		202
	Welding not as specified	105	
	Sheet metal fractured, dented, torn, wrinkled, or otherwise malformed	106	
	Any component misplaced or not in proper alignment		203
	Sharp edges, burrs, or slivers		204
	Any loose particles inside tank (as noted when shaking)		205
	Any operation omitted or not performed as specified	107	
	Any functioning component that is not inoperative or will not operate as intended	108	
	Castings with prominent parting lines due to improper molding	109	
	Cracks	110	
	Not clean		206
	Projections or metal breakout from mold	111	
Heater assembly	Spot welds broken, missing, or cratered due to metal burning	112	
	Sloppy out of tolerance fit	113	
Welding	Any weld missing, incomplete, cracked, fractured, has burn holes, or fusion incomplete	114	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (cont'd)			
Welding (cont'd)	Not smooth, scale or flux deposits, excess metal not removed		207
	Blowholes, hard spots, cold shuts, or porosity	115	
	Not free from spatter		208
Threaded components	Threads broken, stripped, fractured, or missing	116	
	Components not securely tightened		209
	Excess sealing compound not removed		210
Identification marking	Missing, incomplete, or not as specified	117	
	Illegible, incomplete, incorrect, or wrong location		211
Data nameplates	Omitted or not as specified		212
	Information incomplete or not legible		213
	Not located so as to be visible to operator		214

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions identified with an asterisk on the drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of heaters. The sample unit shall be one completely fabricated heater. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.5 End item testing. The end items shall be tested for the characteristics listed below. The lot size shall be expressed in units of space heaters. The sample unit shall be one heater. The service life test shall be the last test performed. The inspection level shall be S-1. Any test failure shall be cause for rejection of the lot.

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TABLE III. End item tests

Characteristic	Requirement paragraph	Test method
Fan capacity	3.3.1.1	4.5.10
Service life	3.4	4.5.6
Heater output	3.4.1	4.5.9
Seepage and burn-off	3.4.2	4.5.4
Tilted operation	3.4.3	4.5.5
Fuel control valve assembly flow rate	3.4.5	4.5.2
Constant level valve	3.4.5	4.5.2
Smoke	3.4.6	4.5.7
Combustion efficiency	3.4.7	4.5.3 and 4.5.8

4.4.6 Codes and standards examination. Certifications, certified test reports, or listing marks for codes and standards, as applicable, submitted in accordance with 3.2, shall be examined and validated as proof of compliance. Prior to Government approval of the first shipment, the contractor shall submit to the contracting officer or his authorized representative for approval, certification that the space heaters he proposes to furnish under this specification meet the applicable requirements of UL as follows:

4.4.6.1 Underwriters' Laboratories, Inc. (UL). Acceptable evidence of meeting the applicable requirements of the Underwriters' Laboratories standards specified in 3.2 shall be the UL label, a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the space heaters have been tested and conform to the applicable requirements of the specified UL standards.

4.4.7 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (including matchmarking, where required)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified Preservatives (where required) missing or not as specified 1/
Workmanship	Inadequate application of components, such as: loose strapping, inadequate stapling, improper taping, or incomplete sealing or closure of flap Bulged or distorted container

1/ Contents of one shipping container from the sample for each lot shall be examined for this condition.

4.5 Methods of inspection.

4.5.1 Conditions. Unless otherwise specified (see 6.2), tests on the heater shall be conducted utilizing all components except the auxiliary fan. All operating tests on the heater shall be conducted while burning Continental U.S. diesel fuel conforming to VV-F-800, grade DF-2. No braces, tiedown devices, or other items that are not inherently a part of the operating heater shall be employed. No alignment of components, replacement of components, or other servicing shall be performed on the first article or production heaters after testing of the heater has been initiated. Tests on the auxiliary fan shall be made without the heater. Efficiency tests shall be conducted with the heater, stove pipes, elbow, and tee installed as shown on the applicable drawing, and tested in accordance with installation test of UL 896. The heater shall be in operation at the prescribed fuel rate with the draft regulator adjusted to 0.060 ± 0.005 inches of water column for a period not less than 15 minutes to allow for the heater to stabilize. Three readings shall be taken at intervals of 15 minutes. The average reading shall be taken for the final determination.

4.5.2 Siphon tube and fuel control valve assembly flow rate test. Prior to assembly on the heater, the constant level valve, with the strainer in place and assembled with the siphon tube assembly and the pot burner fuel line, shall be tested for flow rate. The valve control knob shall be turned to the maximum flow setting and the fuel allowed to flow to purge air from the system. After purging, the fuel shall flow freely through the valve at each setting on the valve. The flow of fuel shall stop when the valve control knob is turned to the "off" position. Inability of the fuel to flow freely when the valve is set at the open positions or to stop flowing when the valve is turned to the "off" position shall constitute a test failure. Failure to comply with the siphon tube and fuel control valve assembly requirements of 3.4.5 shall be cause for rejection of the lot.

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4.5.3 Efficiency. Test the heater for efficiency and compute the output as follows:

a. Assemble the heater per Drawing 5-13-2851. Use eight lengths of stove pipe and the rain cap. The assembly shall be level.

b. Use the constant level valve with strainer, previously tested in accordance with 4.5.2.

c. Use the siphon tube assembly to start the flow of fuel to the constant level valve. Adjust the flow to the maximum setting.

d. Determine the combustion efficiency and output in accordance with UL Standard 896 and 4.5.8.

e. Determine if the flow of fuel ceases, when the valve is shut off, and when the pot is flooded.

f. Readings and calculations shall be reported on the Data and Report Sheet (see table IV).

Inability to start or to operate as required on medium flow, nonconformance to 3.4.7, evidence of oil leaks, or inability to shut off shall constitute failure of this test.

4.5.4 Seepage and burn-off test. Test the heater in accordance test No. 11 of UL Standard 896. Nonconformance to 3.4.2 shall constitute failure of this test.

4.5.5 Tilted operation test. Tilt the heater in four different positions, 90 degrees apart, and at a slope of 2 degrees from the horizontal. Start and operate in each position for a period of 30 minutes. Inability of the heater to start or operate shall constitute failure of this test (see 3.4.3).

4.5.6 Service life test. With the fan (when furnished) installed not more than 3 feet above the heater, operate the heater and fan for 20 hours at high fire. No maintenance is permitted. A military rectangular 5-gallon fuel can, a siphon, and pump shall be used throughout this test. Successful completion of this test shall constitute compliance with the 3000-hour minimum operating life requirement, 300-hour prior to cleaning requirement, and 600-hour prior to maintenance requirement (see 3.4).

4.5.6.1 Failure criteria. One or more of the following shall constitute failure of this test:

a. Inability of the heater to complete 20 hours without maintenance.

b. Incorrect function of the constant-level valve, burn through, or cracking or breaking of metal, warping affecting operation, or a variation of more than 15 percent in per-hour fuel consumption.

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c. Incorrect function of fan components.

d. Dripping of oil from motor sealed bearings.

e. Any operating difficulty, malfunction, or permanent deformation affecting operation or breakage.

4.5.7 Smoke reading test. Five smoke readings shall be taken by the smoke measuring instrument specified in ASTM D 2156, when the heater is operating as specified in 4.5.3.

4.5.8 Combustion efficiency test.

4.5.8.1 Carbon dioxide. The carbon dioxide samples shall be taken in accordance with the installation test of UL Standard 896 using a CO_2 analyzer.

4.5.8.2 Temperature readings. The ambient and the flue gas temperature shall be taken in accordance with the installation test of UL Standard 896.

4.5.8.3 Combustion efficiency. The following formula shall be used for determining the combustion efficiency:

$$E = 1 - \left[\frac{3171 + 2.69\ Tf + \left[\frac{216.2 + 0.77}{CO_2} \right] (Tf - Ta) - 5.38\ Ta}{HHV} \right]$$

E = Combustion efficiency (percent)
Tf = Flue gas temperature (^{o}C)
Ta = Ambient temperature (^{o}C)
CO_2 = Carbon dioxide (percent)
HHV = High heat value of the fuel (Joules per gram)

4.5.9 Heater output test. The fuel as specified in 3.4.1 shall be assumed to have a gross heating value of 19,750 BTUs per pound. The combustion efficiency (see 3.4.7) as determined by 4.5.8.3 shall be used to determine the output as follows:

Heater output (BTU/hr) = Efficiency x Input (BTU/hr)

4.5.10 Fan capacity test. The fans shall be tested in accordance with ASHRAE 51-1985/AMCA 210-85 to determine fan capacity for 0 pressure. Non-conformance with the 650 CFMs specified on the drawings shall constitute failure of this test.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Disassembly and matchmarking. The heater components shall be disassembled as required to accomplish the unit packing specified in 5.1.1.1.8. Disassembled components shall be matchmarked to facilitate reassembly. Matchmarking information shall be on cloth tags conforming to UU-T-81, type A, attached to mating parts.

5.1.1.1 Level A.

5.1.1.1.1 Preservatives. Cleaning, application and type of preservation, and method of preservation shall be in accordance with MIL-P-116 as specified herein.

5.1.1.1.2 Unprotected surfaces. Unpainted and untreated metal surfaces of the heater and components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.1.1.1.3 Fuel system and burner. Interior surfaces of the siphon tube, the valve, and the fuel lines shall be coated with preservative conforming to MIL-L-21260, type I, grade 30. Excess preservative shall be drained.

5.1.1.1.4 Auxiliary fan assembly. The box containing the fan assembly shall be waterproof sealed with tape in accordance with the appendix to PPP-B-636.

5.1.1.1.5 Leveling screws, pins, straps, and other fasteners. Screws, pins, straps, and fasteners shall be coated with type P-7 preservative and preserved in accordance with MIL-P-116, method IC-1.

5.1.1.1.6 Instructional material. Instructional material for each heater shall be preserved together in accordance with MIL-P-116, method IC-1.

5.1.1.1.7 Technical publications. When technical publications are furnished, they shall be individually preserved in accordance with MIL-P-116, method IC-1.

5.1.1.1.8 Unit packing. Heaters without fan and smoke jack assembly shall be packed in accordance with table V and figures 1 and 2, level A. Heaters with fan and smoke jack assembly shall be packed in accordance with table VI and figures 3 and 4, level A.

5.1.1.2 Level B. Preservation shall be as specified for level A except that unit packing for heaters without fan and smoke jack assembly shall be in accordance with table V and figures 1 and 2, level B. Unit packing for heaters with fan and smoke jack assembly shall be in accordance with table VI and figures 3 and 4, level B.

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5.1.1.3 Level C. The heater components, technical publications, and instructional material shall be preserved to afford protection against deterioration and damage from the supplier to the initial destination. Unit packing for heaters without fan and smoke jack assembly shall be in accordance with table V and figures 1 and 2, level C. Unit packing for heaters with fan and smoke jack assembly shall be in accordance with table VI and figures 3 and 4, level C.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A packing. Heaters without fan and smoke jack assembly shall be packed in accordance with table V and figures 1 and 2, level A. Heaters with fan and smoke jack assembly shall be packed in accordance with table VI and figures 3 and 4, level A. When technical publications are furnished, they shall be packed with the respective item.

5.2.2 Level B packing. Heaters without fan and smoke jack assembly shall be packed in accordance with table V and figures 1 and 2, level B. Heaters with fan and smoke jack assembly shall be packed in accordance with table VI and figures 3 and 4, level B. When technical publications are furnished, they shall be packed with the respective item.

5.2.3 Level C packing. Heaters without fan and smoke jack assembly shall be packed in accordance with table V and figures 1 and 2, level C. Heaters with fan and smoke jack assembly shall be packed in accordance with table VI and figures 3 and 4, level C. When technical publications are furnished, they shall be packed with the respective item.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The heaters are intended for use primarily for heating prefabricated and temporary-type buildings.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).

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d. Required data will be made available unless otherwise specified (see 3.4).
e. When a smoke jack assembly is required (see 3.4).
f. When a fan assembly is required (see 3.4).
g. When a government approved manual number is to be supplied with data nameplate (see 3.6).
h. Conditions of test, if different (see 4.5.1).
i. Level of preservation and packing required (see 5.1 and 5.2). Level B preservation and packing is intended to provide limited but economical protection and should be specified only when it is determined the heaters will be held in covered storage.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Supplementary quality assurance provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Supplementary quality assurance provisions (SQAP). A SQAP is a contractural requirement that supplements section 4 of the document. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Subject term (key word) listing.

Fuel burning space heater
Oil fired air heater
Oil fired unit-heater
Radiant space heater
Space heater

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - ME
Air Force - 84
DLA - CS

User activity:

Navy - MC

Preparing activity:

Army - GL

(Project 4520-0354)

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TABLE IV. Oil-heater rating test
Data and Report Sheet

Manufacturer's test No._____

Oil heater _____
(Make) (Model, type, or no.) (Number and size of burners)

Fuel used for test _____
(Brand) (API gravity No. at 60°F)

Date of test _____ Tested by _____

Test data	Three sets of readings at equal intervals not less than 20 minutes apart.			Average
1. Draft _____(in. of water)	____	____	____	_____
2. Room temperature _____(°F)	____	____	____	_____
3. Smoke-meter readings _____(spot # ___)	____	____	____	_____
4. Fuel-temperature readings _____(°F)	____	____	____	_____
5. Time intervals used for fuel readings __	____	____	____	_____
6. Fuel input readings (specify units used)	____	____	____	_____
7. Fuel-oil input rate _____(lb/hr avg)___				_____
8. Gross heating value of oil _____(BTU/lb)__				_____
9. Gross heat input _____(BTU/hr)__				_____
10. Flue-gas temperature _____(°F)	____	____	____	_____
11. Flue-gas temp. rise above room temp (°F)	____	____	____	_____
12. CO_2 in dry flue gas _____(%)_____	____	____	____	_____
13. CO _____ (%)			____	_____
14. Efficiency for complete combustion in accordance with the efficiency formula in 4.5.8.3. (%)			____	_____
15. Heat output in accordance with the heater output formula in 4.5.9 (BTU/hr)			____	_____
16. Fuel-input rate for low-fire smoke test _____ (lb/hr)			____	_____
17. Smoke-meter readings at 20 minute intervals (low-fire test) _____ (spot #__)	____	____	____	_____

We hereby certify that the above is a true and accurate copy of the data obtained when the above heater was tested in accordance with this document.

For _____
Address _____
Date _____ By _____

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TABLE V. Preservation and packing material for heaters without fan and smoke jack assembly

FIND NO.	IDENTIFICATION	DESCRIPTION	LEVEL A	LEVEL B	LEVEL C
1	PPP-B-601, overseas type, style A, type 3 load (see Note 1)	Box, Plywood, Cleated	1		
1	PPP-B-591, class II, style A (see Note 1)	Box, Fiberboard, Cleated		1	
2	PPP-B-636, V3c, style RSC	Box, Fiberboard	1	1	
2	PPP-B-636, type CF, class domestic, variety SW, grade 275, style RSC	Box, Fiberboard			.1
3	PPP-F-320, type CF, class domestic, variety SW, grade 125	Fiberboard	1	1	1
4	Base, side members	Lumber, softwood 1 x 6 nominal	2	2	2
5	Base, cross members	Lumber, softwood 1 x 3 nominal	2	2	2
6	FF-N-105, type II, style 4, table Vb	Nail, 6d	AR	AR	AR
7	MS27183-11, .312 IDx.734 ODx.065 THK	Washer, flat-round, steel, cad plated, general purpose	4	4	4
8	MS35338-45, 5/16 nominal size	Washer, lockspring, helical, regular series, stl, cad plated	4	4	4
9	MS35206-301, 5/16-18UNC-2A x 1-3/4 long	Screw, mach, pan hd, cross-recessed, cs, cad plated	4	4	4
10	ASTM A641, 2 mm dia, class 1, soft	Wire, light, soft tensile, light zinc coated	AR	AR	AR

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TABLE V. Preservation and packing material for heaters without fan and smoke jack assembly (cont'd)

FIND NO.	IDENTIFICATION	DESCRIPTION	LEVEL A	LEVEL B	LEVEL C
11	PPP-F-320, Type CF, class domestic, variety SW, grade 125	Fiberboard	AR	AR	AR
12	PPP-F-320, Type CF, class domestic, variety TW, grade 1100	Fiberboard	1	1	1
13	PPP-F-320, Type CF, class domestic, variety TW, grade 1100	Fiberboard	1	1	1
14	ASTM D 3953, type 1 or type 2, coated finish	Steel strapping, flat, .035 x 3/4 W	AR	AR	AR
15	PPP-T-60, type IV, class I	Tape, pressure sensitive	AR	AR	AR
16	Blocking	Lumber, softwood 1 x 3 nominal	2	2	2
17	ASTM D 3953, type 1, coated finish	Steel strapping, flat, .023 x 1/2 W	AR	AR	AR
18	Blocking	Lumber, softwood 2 x 2 nominal x 3LG	2	2	2
19	A-A-1898, grade II, class A (any style, any size)	Cushioning	AR	AR	AR
20	PPP-T-97, type III	Tape, pressure sensitive 3/4 W	AR	AR	AR

NOTES:
1. Top to be inverted.
2. AR denotes "as required."

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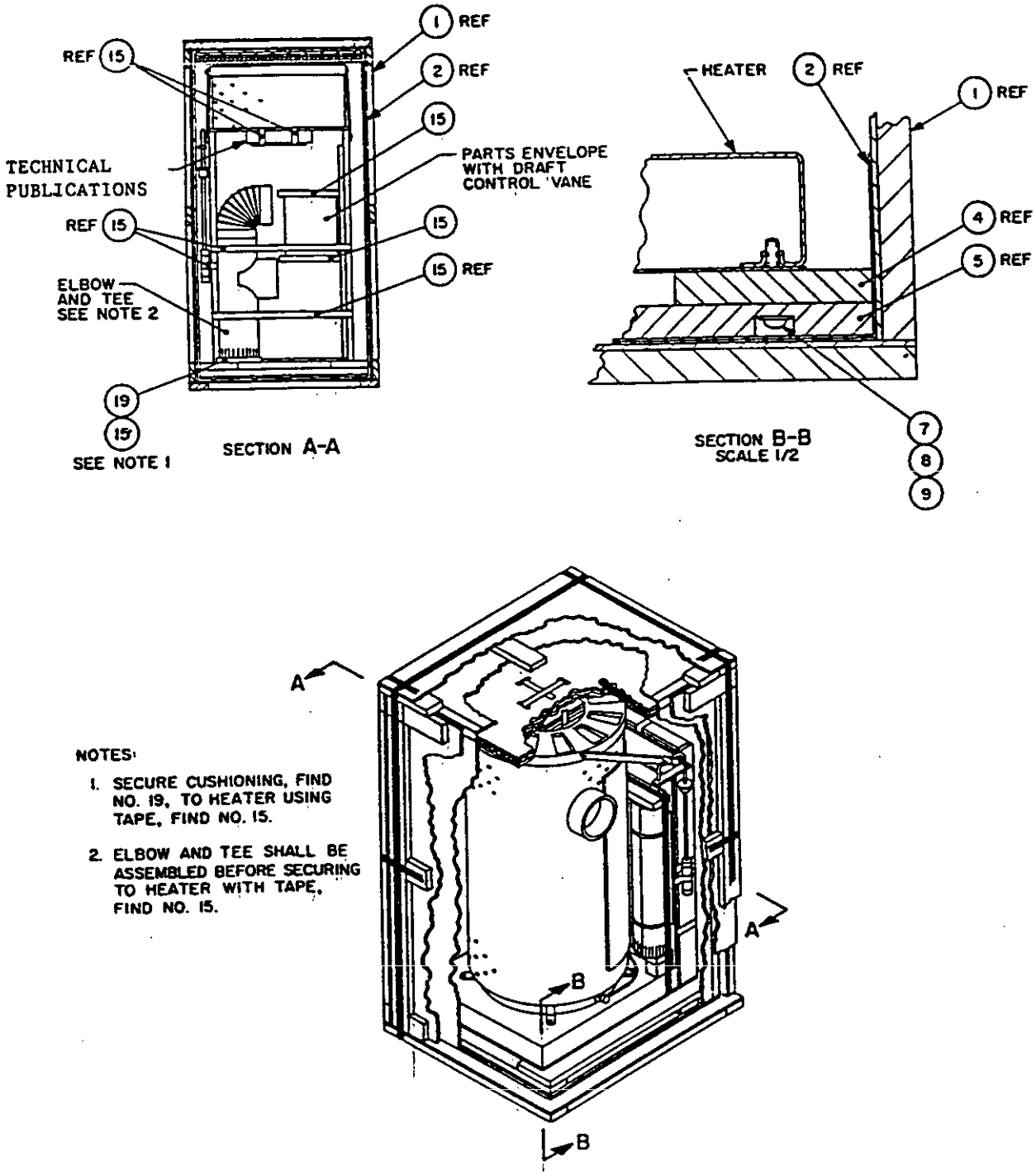


FIGURE 1. Packing arrangement and details for heaters without fan and smoke jack assembly.

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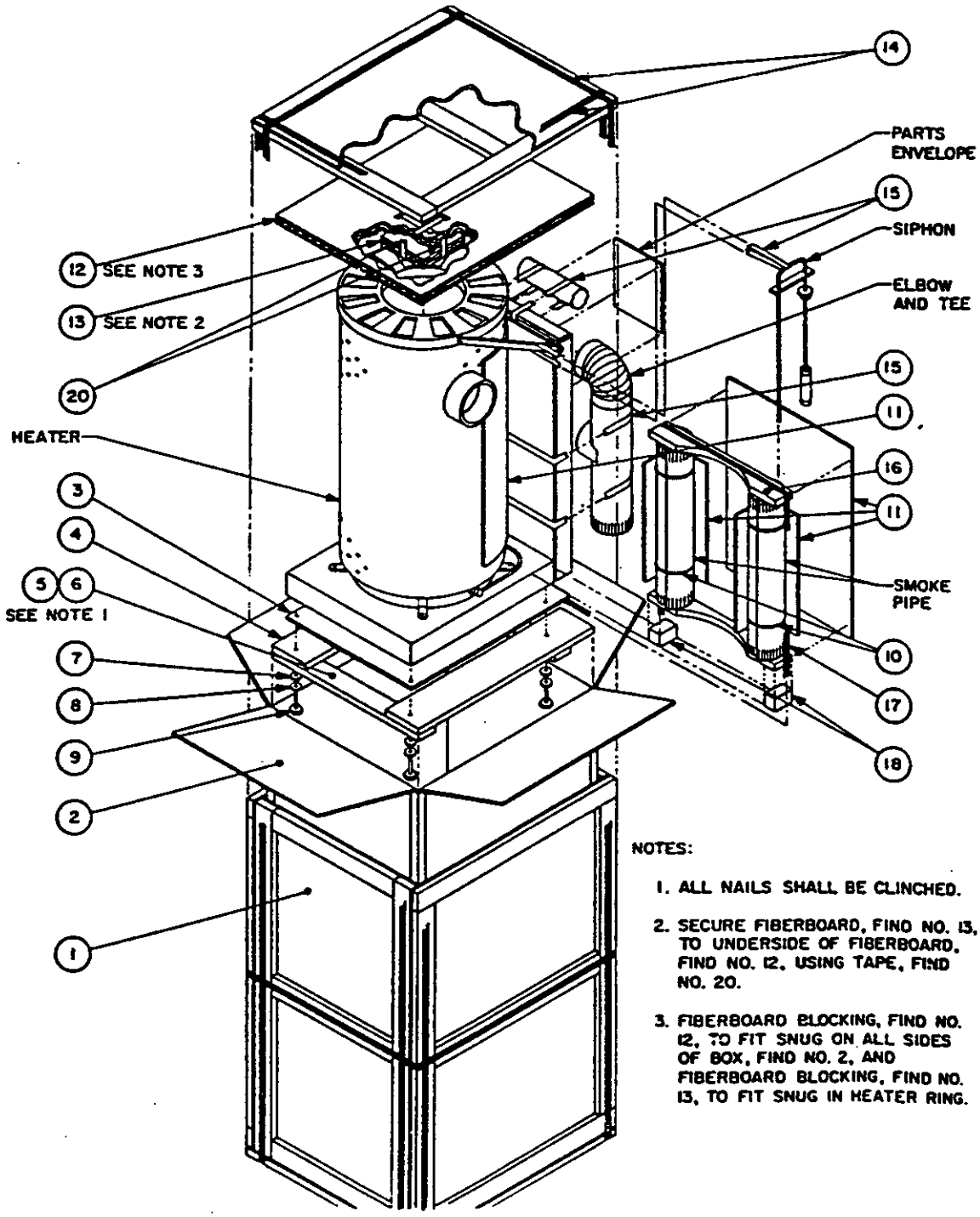


FIGURE 2. Packing assembly for heaters without fan and smoke jack assembly.

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TABLE VI. Preservation and packing material for heaters with fan and smoke jack assembly

FIND NO.	IDENTIFICATION	DESCRIPTION	LEVEL A	LEVEL B	LEVEL C
1	PPP-B-601, overseas type, style A, type 3 load (see Note 1)	Box, Plywood, Cleated	1		
1	PPP-B-591, class II, style A (see Note 1)	Box, Fiberboard, Cleated		1	1
2	PPP-B-636, V3c, style RSC	Box, Fiberboard	1	1	
2	PPP-B-636, type CF, class domestic, variety SW, grade 275, style RSC	Box, Fiberboard			.1
3	PPP-F-320, type CF, class domestic, variety SW, grade 125	Fiberboard	AR	AR	AR
4	Base, side members	Lumber, softwood, 1 x 3 nominal	2	2	2
5	Base, front cross member	Lumber, softwood, 1 x 6 nominal	1	1	1
6	Base, rear cross member	Lumber, softwood, 1 x 3 nominal	1	1	1
7	FF-N-105, type II, style 4, table Vb	Nail, 6d	AR	AR	AR
8	MS27183-11, .312 IDx.734 ODx.065 THK	Washer, flat-round, steel, cad plated, general purpose	4	4	4
9	MS35338-45, 5/16 nominal size	Washer, lockspring, helical, regular series, stl, cad plated	4	4	4
10	MS35206-301, 5/16-18UNC-2A x 1-3/4 long	Screw, mach, pan hd, cross-recessed, cs, cad plated	4	4	4
11	PPP-T-60, type III, class I	Tape, pressure sensitive	AR	AR	AR

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TABLE VI. Preservation and packing material for heaters with fan and smoke jack assembly (cont'd)

FIND NO.	IDENTIFICATION	DESCRIPTION	LEVEL A	LEVEL B	LEVEL C
12	ASTM A641, 2 mm dia., class 1, soft	Wire, light, soft tensile, light zinc coated	AR	AR	AR
13	A-A-1898, grade II, class A (any style, any size)	Cushioning	AR	AR	AR
14	Blocking	Lumber, softwood, 1 x 3 nominal	3	3	3
15	ASTM D 3953, type 1 or 2, coated finish	Steel strapping, flat, .023 x 1/2 W	AR	AR	AR
16	PPP-B-636, W5c	Box, Fiberboard	1		
16	PPP-B-636, type CF, class domestic, variety SW, grade 175	Box, Fiberboard		1	1
17	PPP-F-320, Type CF, class domestic, variety TW, grade 1100	Fiberboard	AR	AR	AR
18	PPP-T-97, type III, 3/4 w	Tape, pressure sensitive	AR	AR	AR
19	ASTM D 3953, type 1 or type 2, coated finish	Steel strapping, flat, .035 x 3/4 W	AR	AR	AR
20	MS24629-48, number 10-24 x 3/4 LG	Screw, tapping - THD cutting, type D, pan hd, cross-recessed, CS, CAD, PLD	2	2	2
21	Blocking	Lumber, softwood, 2 x 2 nominal	AR	AR	AR
22	PPP-B-636, W5c	Box, Fiberboard	1		
22	PPP-B-636, type CF, class domestic, variety SW, grade 175	Box, Fiberboard		1	1

NOTES:
1. Top to be inverted.
2. AR denotes "as required."

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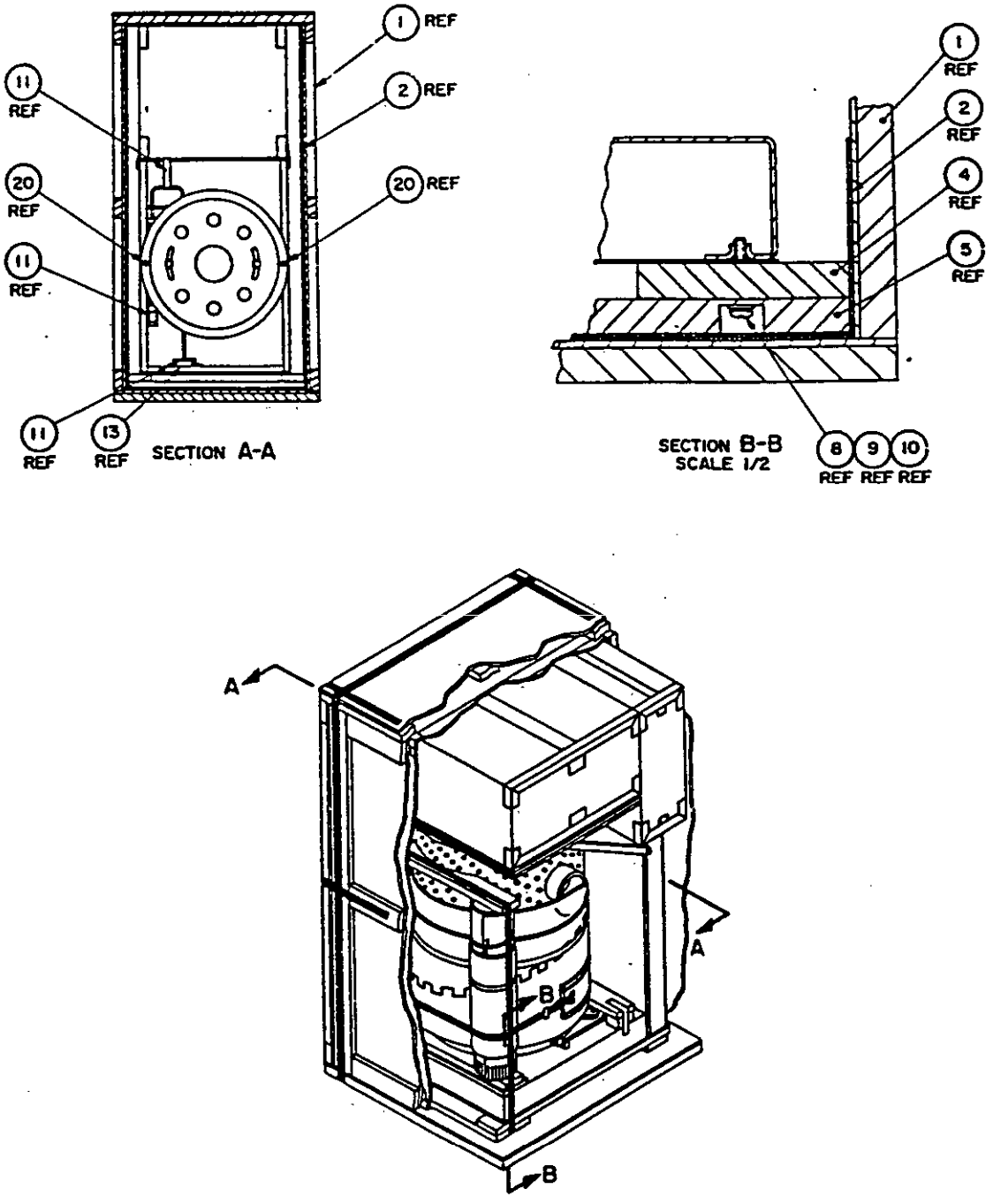


FIGURE 3. Packing arrangement and details for heaters with fan and smoke jack assembly.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-14516F	1993 July 28

3. DOCUMENT TITLE

HEATER, SPACE: OIL FIRED, BARRACKS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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